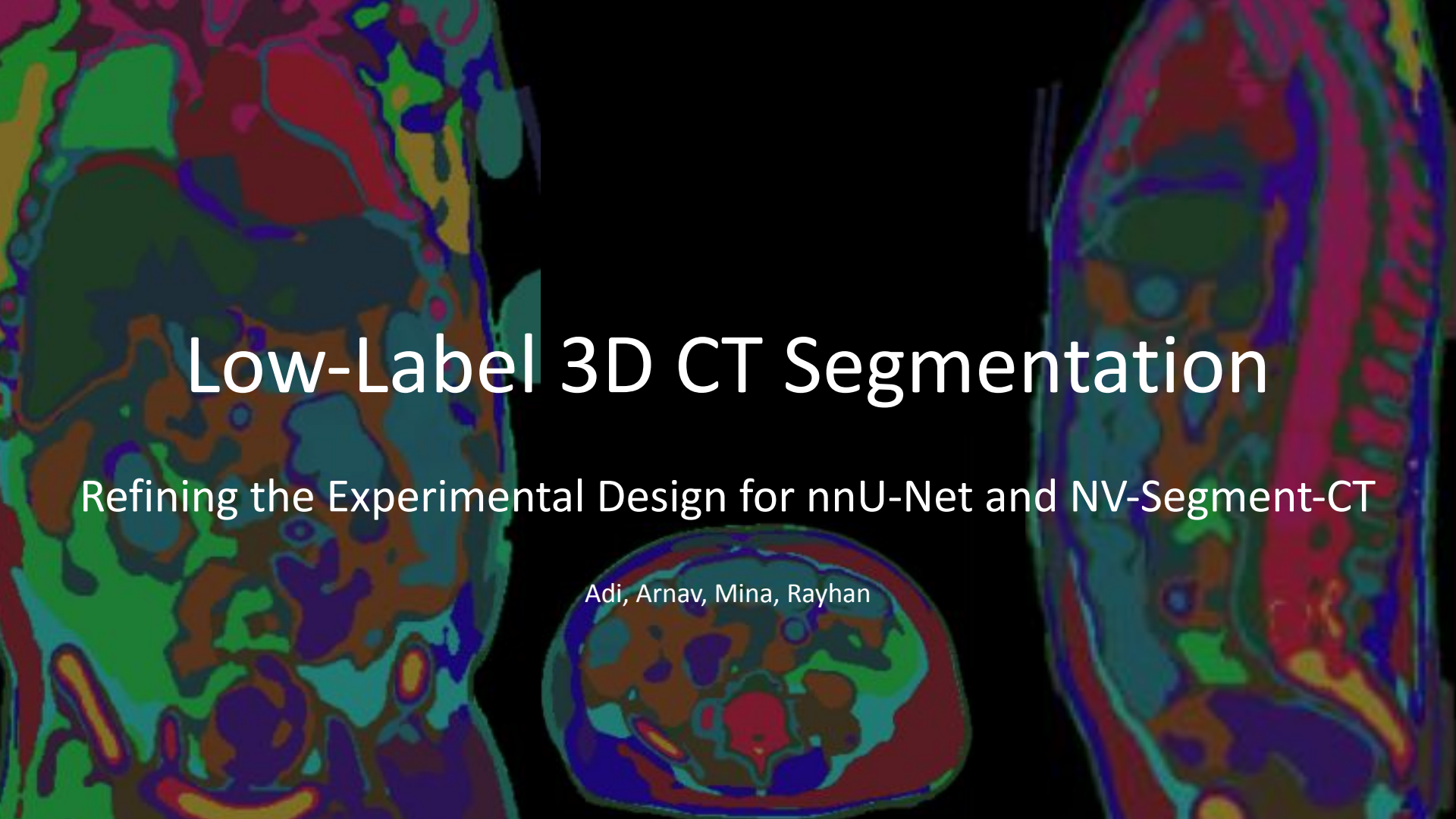


Low-Label 3D CT Segmentation

Refining the Experimental Design for nnU-Net and NV-Segment-CT

Adi, Arnav, Mina, Rayhan

Overview of Aims

Aim 1: Patient-level label scarcity

- Evaluate segmentation performance between nnU-Net and NV-Segment-CT on fully labeled patient subsets.
- Segmentation masks will be benchmarked against ground-truth labels using the volumetric Dice Similarity Coefficient (DSC) for spatial overlap and the 95th-percentile Hausdorff Distance (HD95) for contour boundary fidelity

Aim 2: Distributed annotation budget

- Evaluate segmentation performance when nnU-Net and NV-Segment-CT are trained on a set of partially labeled patients
- We need to take OAR size into consideration when assessing performance and adapt both models to take in partial annotations.

Aim 3: Value of pretraining

- Determine when pre-training provides an advantage over training fully from scratch
- Will measure and evaluate the difference in between NV-Segment-CT and nnU-Net under every scarcity condition, and look for where pre-training helps most: at which budgets, and for which anatomical structures

Overview of Data Preprocessing

1) Input data: SegRap2023
   a) 120 patients
   b) Paired ncCT + ceCT scans
   c) 45 OARs + 2 GTVs
2) Data preparation
   a) Align paired ncCT + ceCT scans
   b) Standardize label mappings for both NV-Segment-CT and nnU-Net
3) nnU-Net standardization
   a) Create 2-channel 3D inputs and separate OAR + GTV training
   b) Analyze automatic nnU-Net preprocessing choices
   c) Standardize CT scan intensities across each input channel
4) NV-Segment-CT standardization
   a) Align PAIRED ncCT + ceCT volumes
   b) Modify the pretrained input layer to accept two channels and outputs to represent SegRap dataset
   c) Resample the 0.55 × 0.55 × 3.0 mm scans to the foundation model's 1.5 mm isotropic spacing
   d) Remap SegRap classes to NV-Segment-CT outputs

nnU-Net

Fully Annotated Patients

nnU-Net

Partially Annotated Patients

NV-Segment-CT

Fully Annotated Patients

NV-Segment-CT

Partially Annotated Patients

Overview of Fine-Tuning Process

- Preprocessed 5-patient pilot cohort into 1.5 mm isotropic tensors
- Adapted NV-Segment-CT foundation model to accept 2-channel inputs and output 45 OAR classes
- Completed 50-epoch GPU training and compared pretrained/scratch NV-Segment-CT models
- Computed DSC, combined soft dice and BCE loss

Biggest Challenge: SegRap Labels vs. NV-Segment-CT Classes



47 SegRap2023 targets (45 OARs + 2 GTVs)

Direct match (5)

Brain, esophagus, spinal cord, thyroid, trachea

Sub-region (10)

Brainstem, hippocampus, larynx, oral cavity, temporal lobes

Nearest: skull (18)

Cochlea, mandible, middle ear, mastoid, TMJ

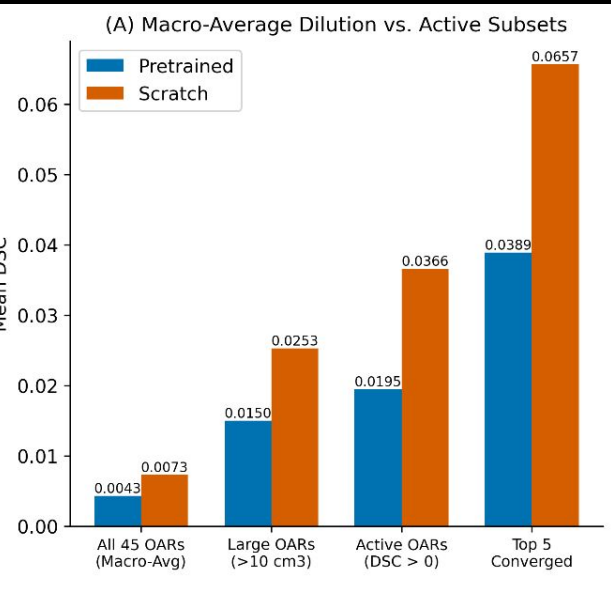
No analog (14)

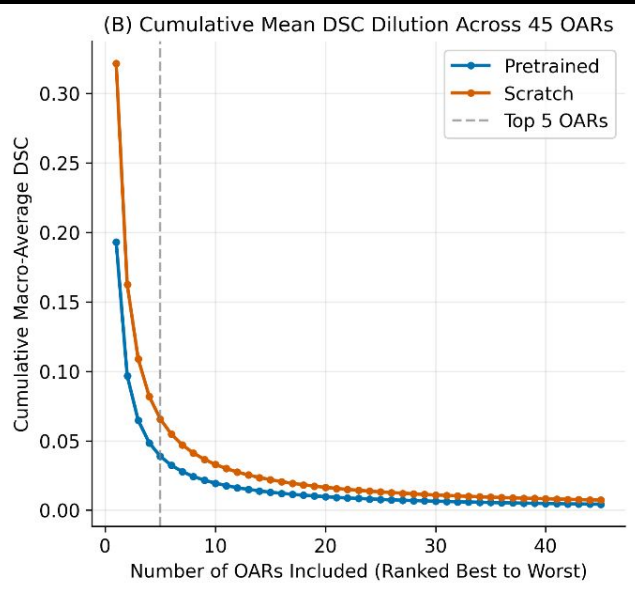
Chiasm, eyes, lens, optic nerves, parotids, pituitary, GTVs

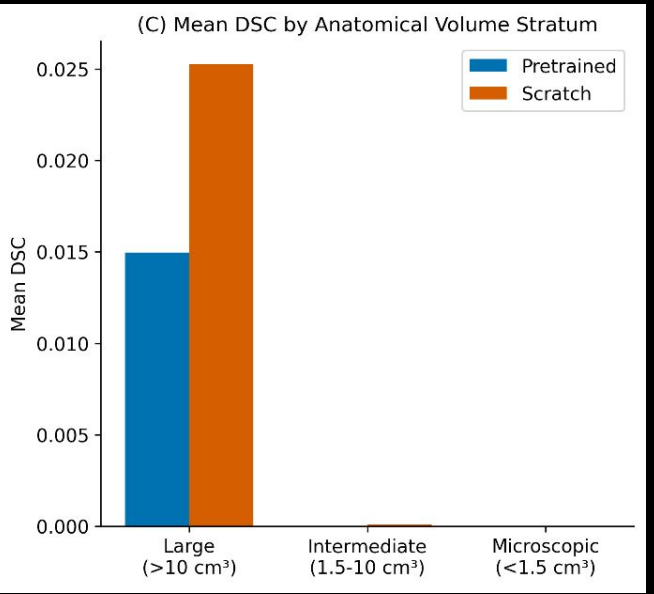
Takeaway: 89% of targets have no direct pretrained class

Subsequent challenge: Isotropic resampling may distort/erase smaller structures.

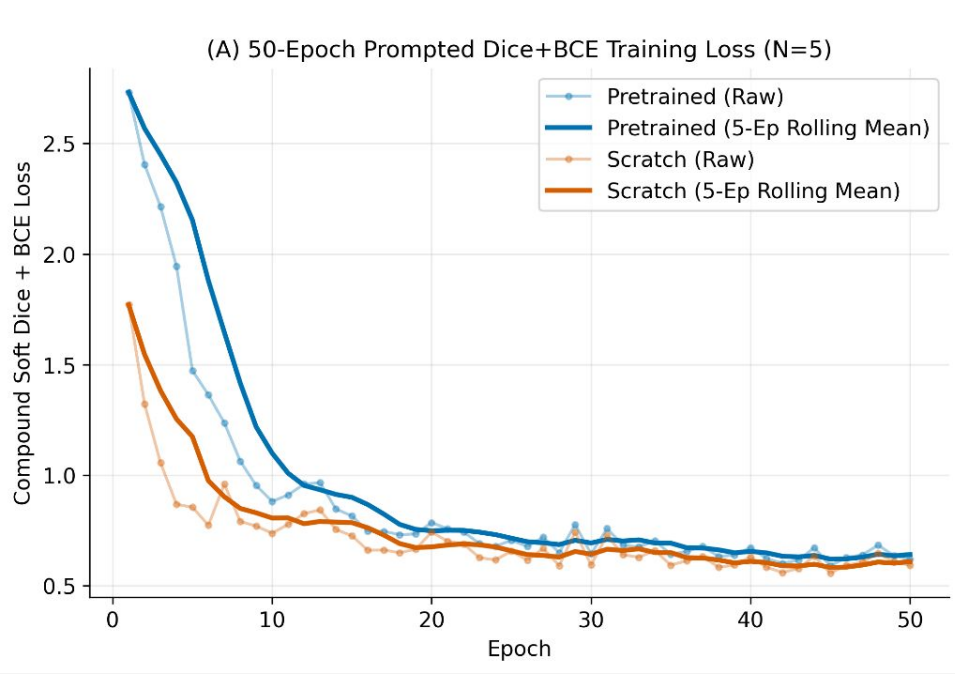
Aim 1B Updates: NV-Segment-CT with **Patient-Level Label Scarcity**

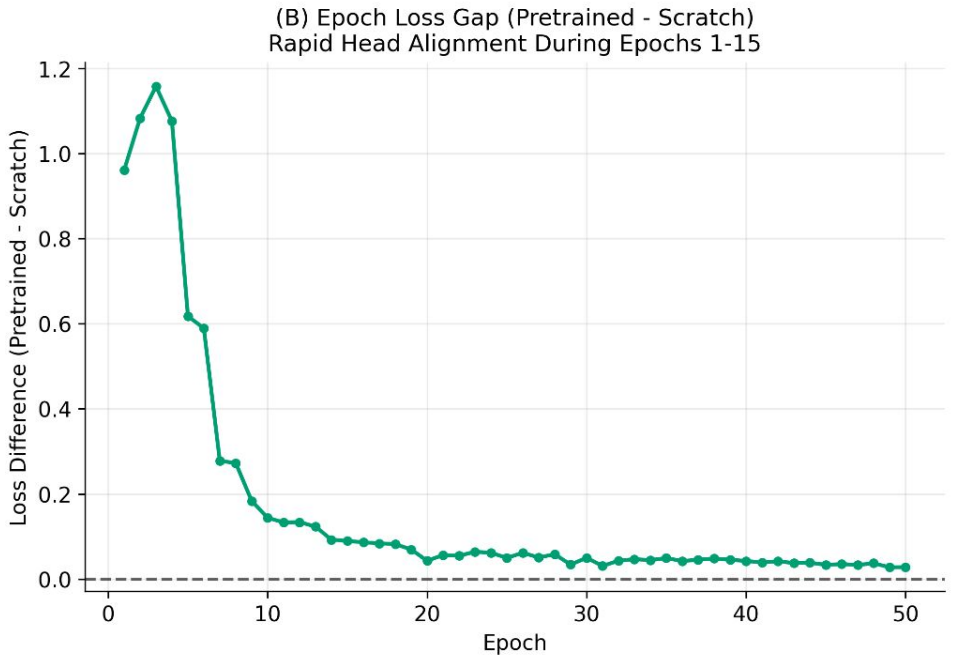




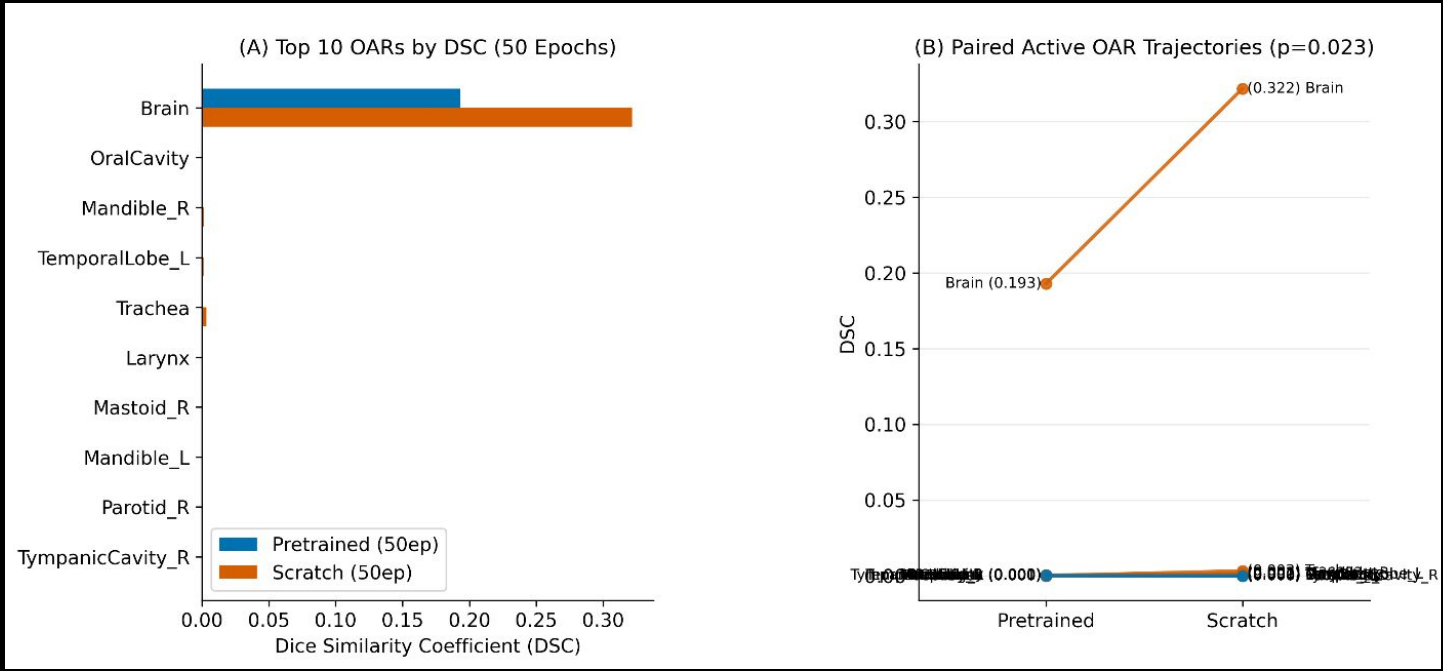


Aim 1B Updates: NV-Segment-CT with **Patient-Level Label Scarcity**

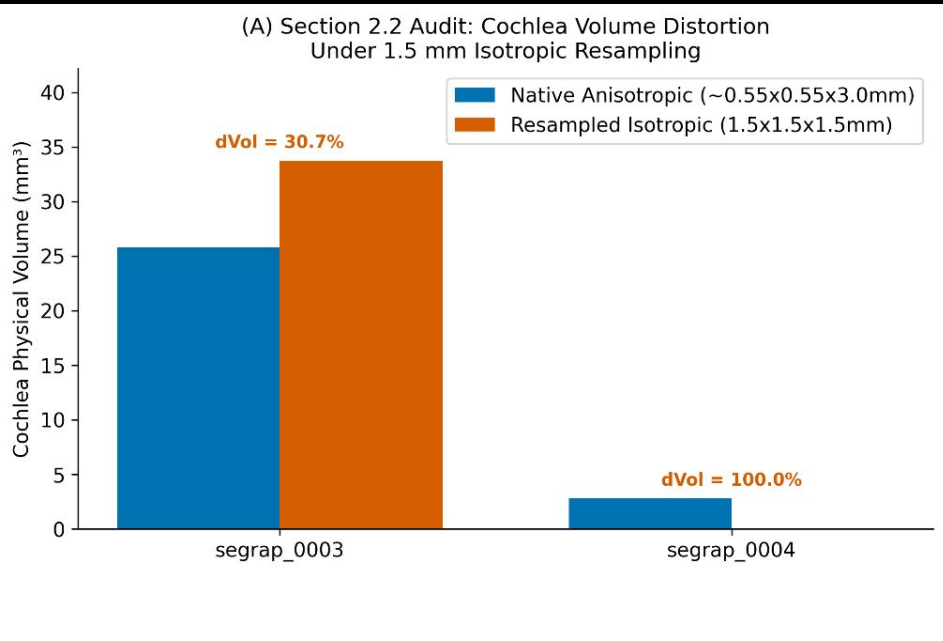


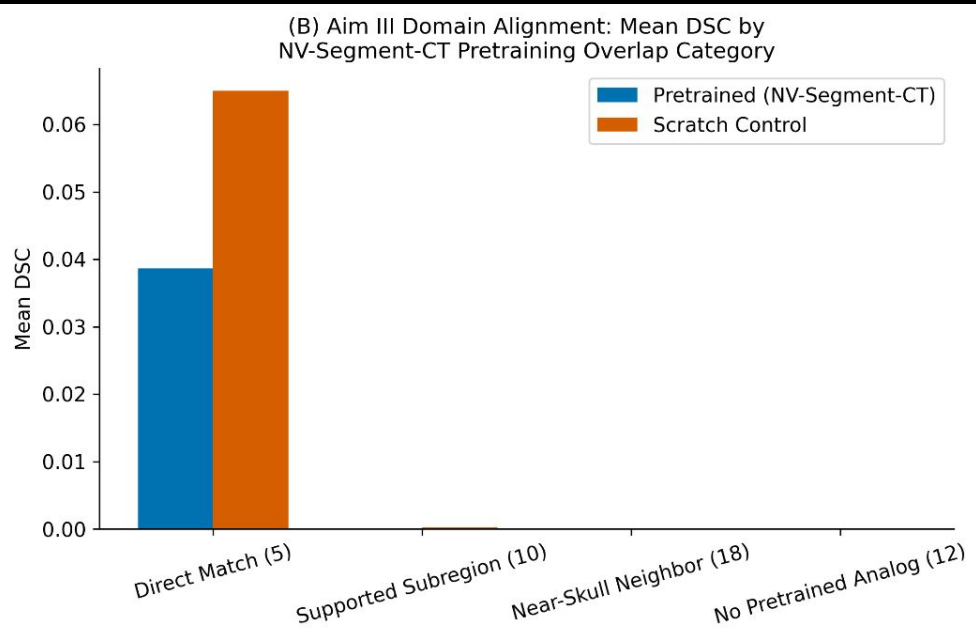


Aim 1B Updates: NV-Segment-CT with **Patient-Level Label Scarcity**



Aim 1B Updates: NV-Segment-CT with **Patient-Level Label Scarcity**





Next Steps

- Set up and implement other baseline tests for nnU-Net and NV-Segment-CT once members have access to Expanse
- Specifically for NV-Segment-CT with fully annotated patients:
  - Targeted foreground patch oversampling (e.g. localized high-resolution ROI cropping) to fix extreme class imbalance
  - Check if unfreezing weights improves performance without overfitting/catastrophic forgetting
  - Increase N

Questions?

